ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2015

BIOLOGY **9648/03**

Applications Paper and Planning Question **28 Aug 2015**

PAPER 3 **2 hours**

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, subject class, form class and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE	
1	
2	
3	
4	
5	
TOTAL	**72**

This document consists of **14** printed pages.

For Examiner's Use

1. Before the emergence of more advanced technologies, cDNA libraries have been used to study the genetic changes involved in cancer development (Fig. 1.1).

Normal cells and tumour cells are obtained from the same patient. Reverse transcription is carried out on mRNA isolated from the tumour cell with the use of primers consisting of repeats of thymine (TTTTT), forming single-stranded cDNA. These cDNA are then labelled with fluorescent dyes. Normal cell mRNA and tumour cell cDNA are allowed to hybridise; the resulting double-stranded hybrid molecules and remaining single-stranded mRNA are discarded. Non-hybridised cDNA (also known as subtracted cDNA) are used to form a subtracted cDNA library, or are used as probes to isolate genes from an existing DNA library.

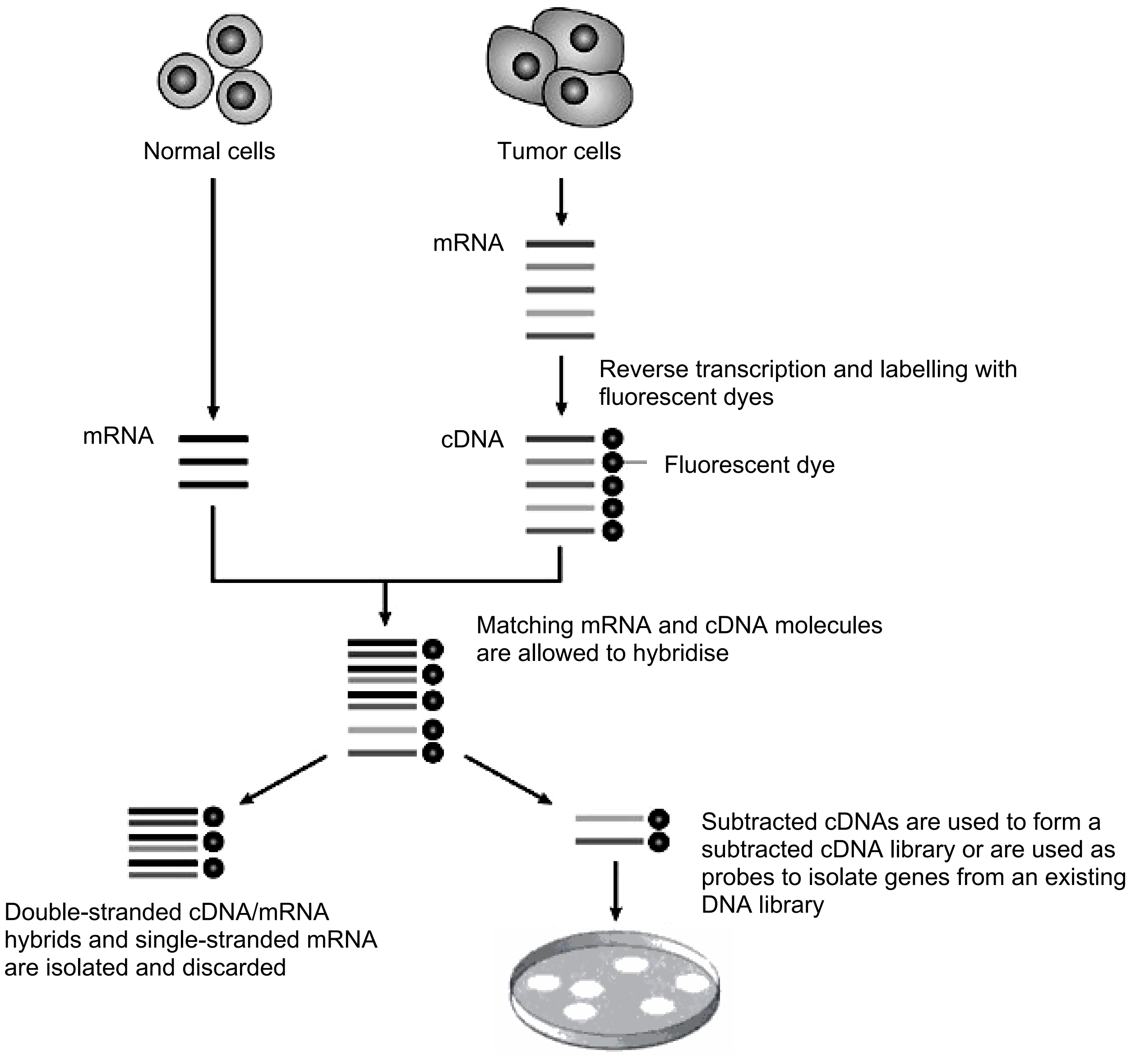

Fig. 1.1

(a) (i) A scientist conducting this procedure was concerned that the tumour cell sample may be contaminated with bacteria. Explain why the reverse transcription of bacterial mRNA will not occur.

[2]

(ii) Describe the steps needed to form a cDNA library using subtracted cDNA and bacterial plasmids.

[5]

(iii) Explain how the subtracted cDNA library can be used to study cancer cells.

[3]

(iv) Suggest why the procedure described in Fig. 1.1 may not be able to detect the presence of mutated tumour suppressor genes.

[2]

A DNA library can also be formed using other vectors and host cells, as shown in Table 1.1. The structure of one such vector, the yeast artificial chromosome (YAC), is shown in Fig. 1.2.

Table 1.1

Vector	Host cell	Transgene capacity
Plasmid	*E. coli*	10 kilobases
Lambda bacteriophage	*E. coli*	20 kilobases
Bacterial artificial chromosome (BAC)	*E. coli*	300 kilobases
Yeast artificial chromosome (YAC)	Yeast cells	2.5 megabases
Human artificial chromosome (HAC)	Human cells	> 2.5 megabases

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Fig. 1.2

(b) (i) Describe **two** advantages of using YAC with yeast as host cells over using plasmids with *E. coli* as host cells, for the large-scale production of eukaryotic proteins.

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(ii) State the role of the multiple cloning site (MCS) found in YAC.

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.... [1]

[Total: 15m]

For Examiner's Use

2. The first genetic maps, constructed in the early decades of the 20th century for organisms such as the fruit fly, used genes as markers. However with advancement in the understanding of genetics and molecular biology, RFLPs have been favoured as genetic markers in constructing the genetic map.

(a) Explain the advantages of RFLPs as genetic markers over genes in the construction of a genetic map.

[4]

(b) In the creation of a genetic map, two RFLP loci on chromosome one are identified in fruit fly. A female heterozygous for both RFLP loci was mated with a male. The RFLP banding patterns of the mating parents as well as the F_1 offspring are shown in Fig. 2.1

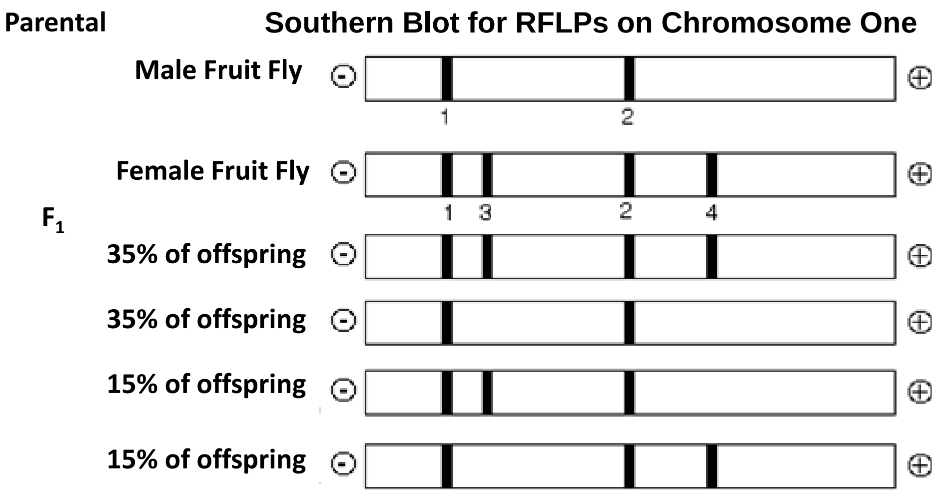

Fig. 2.1

(i) Explain why a female heterozygous for both RFLP loci was used in this experiment.

[2]

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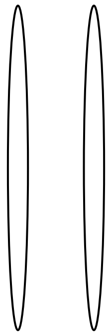
(ii) Draw the positions of the two RFLP loci of the female fruit fly from the parental generation on the pair of homologous chromosomes below, and indicate the genetic distance between the two loci. [1]



(iii) The fragments of DNA were separated using gel electrophoresis in this experiment.

Suggest and explain how increasing the concentration of the agarose during the preparation of the gel would affect the way in which DNA fragments are separated.

[3]

(c) Simple sequence length polymorphism (SSLP) is another type of genetic marker that is commonly used in genetic mapping. SSLPs are repeated sequences in intergenic regions of DNA. The repeated sequences in SSLP are flanked by conserved regions. An example of SSLP is shown in Fig. 2.2.

...gatttctaag**CA**tctttaaatg...
...gatttctaag**CACA**tctttaaatg...
...gatttctaag**CACACA**tctttaaatg...
...gatttctaag**CACACACACA**tctttaaatg...
...gatttctaag**CACACACACACA**tctttaaatg...

Fig. 2.2

Explain one way in which nucleic acid hybridization can be employed to distinguish between SSLP of different lengths without the use of restriction enzymes.

[4]

[Total: 14m]

3. The most common treatment for severe combined immunodeficiency (SCID) is bone marrow transplantation. In the 1990s, gene therapy has been attempted as an alternative to bone marrow transplantation.

(a) **(i)** Explain how mutations to the interleukin-2 receptor gamma (IL2RG) gene result in symptoms exhibited in X-linked SCID patients.

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(ii) Explain one factor that prevents the viral gene delivery method from becoming an effective long - term treatment in SCID patients.

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Recently, it has been proposed that gene therapy can also be used to target cancer cells. Fig. 3.1 shows the process of how a normal white blood cell removed from a lung cancer patient is treated *ex vivo* with a gene coding for P receptors that specifically recognise proteins on cancer cells.

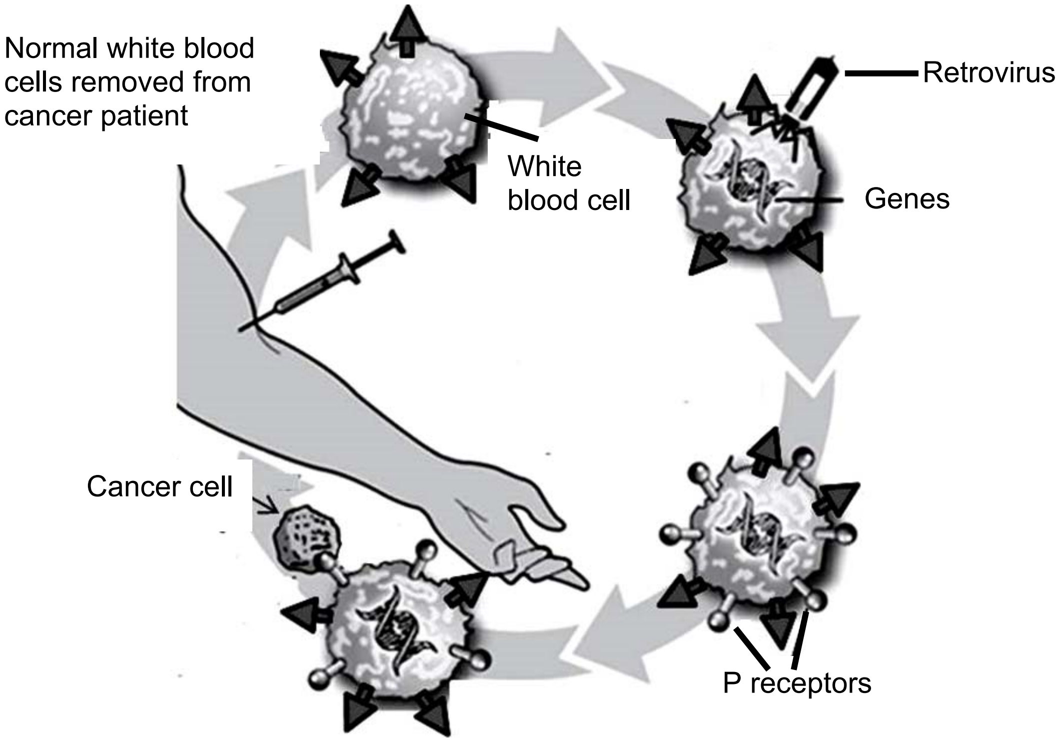

Fig. 3.1

(b) (i) With reference to Fig. 3.1, outline the method of gene therapy used to target cancer cells for degradation.

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(ii) Discuss the ethical considerations for such novel methods of gene therapy in targeting cancer cells.

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____[2]

[Total: 11m]

4. Planning question

The enzyme β-galactosidase hydrolyses the sugar lactose into glucose and galactose that can be absorbed into the blood stream. However, it is not easy to measure β-galactosidase activity using the appearance of glucose and galactose.

Iodine binds to a site on β-galactosidase other than the active site and changes the conformation of the active site.

In order to measure enzyme activity of β-galactosidase, an artificial substrate called o-nitrophenyl-beta-galactopyraniside (ONPG) is used. The enzyme degrades ONPG to produce beta-galactose and o-nitrophenol, a yellow coloured compound. This reaction occurs optimally at pH 5.0. The reaction is stopped by adding sodium carbonate solution to increase the pH to 11.

Using this information and your own knowledge, design an experiment to investigate the effect of increasing the concentration of iodine on the rate of hydrolysis of ONPG by β-galactosidase.

You must use:
- 1.0% β-galactosidase solution,
- 0.3% iodine solution,
- distilled water,
- 5mM ONPG
- pH 5.0 buffer,
- 0.5 M sodium carbonate solution.

You should select from the following apparatus and use appropriate additional apparatus:
- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, glass rods etc.,
- timer e.g. stopwatch or stop clock,
- thermometer,
- colourimeter,
- thermostatically controlled water bath.

Your plan should:
- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12m]

For Examiner's Use

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5. Free-response question

Write your answers to this question on the separate paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

(a) Explain how the findings of the Human Genome Project have been used in the advancement of healthcare and describe the ethical issues related to such use. [8]

(b) Comment on the statement "GMO can make a relevant contribution to solving the problem of food shortage in developing countries". [6]

(c) With reference to a named genetic disease, explain the features of stem cell transplant that allows it to be an effective treatment. [6]

[Total: 20m]